

- 3 (a) series of distinct coloured lines against dark background B1
- (b) photon emitted when electron changes from higher to lower energy B1
- energy of photon proportional to frequency M1
- each line represents photon of specific / distinct energy A1
- only transition between discrete energy levels leads to specific energies
- (c) $\Delta E = 1.5 - 0.85 (= 0.65 \text{ eV or } 1.04 \times 10^{-19} \text{ J})$ C1
- wavelength $= \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.04 \times 10^{-19}} = 1.91 \times 10^{-6} \text{ m}$ M1
- wavelength is longer than visible light / infrared, so not visible A1
- (d) (i) waves passing slits spread B1
- waves from each slit overlap M1
- with phase difference 360° / path difference λ A1
- (ii) angle of λ_1 maximum $= \tan^{-1} \frac{147-73.5}{240} (= 17.027^\circ \text{ or } 0.29718 \text{ rad})$ C1
- $\sin \theta_2 = \sin 17.027^\circ \times \frac{654 \times 10^{-9}}{488 \times 10^{-9}} (\Rightarrow \theta_2 = 23.106^\circ \text{ or } 0.40328 \text{ rad})$ C1
- $\frac{x-73.5}{240} = \tan 23.106^\circ$ M1
- $x = 175.9 \text{ cm}$ A1

4 (a) emission of electrons from a metal surface when surface is irradiated with electromagnetic